M-Series Three-Phase Induction Motors

1. Scope

This manual covers installation and operation of M-Series totally enclosed fan cooled (TEFC) squirrel cage induction motors, frame sizes 160 to 225, for 400 V 50 Hz supply. Maintenance and troubleshooting are covered in M-MT-04. Electrical tests on windings are described in M-ETP-05, which is restricted to qualified electrical technicians.

2. Ratings

Model	Power	Rated current at 400 V	Speed	Efficiency class
M-160L	15 kW	29 A	1460 rpm	IE3
M-200L	30 kW	56 A	1475 rpm	IE3
M-225M	45 kW	82 A	1480 rpm	IE3

2.1 Common data

Item	Value
Insulation class	F (155 °C)
Temperature rise	Class B (80 K)
Degree of protection	IP55
Cooling method	IC411 (shaft-mounted fan)
Duty	S1 continuous
Ambient temperature	-20 to +40 °C
Altitude	up to 1000 m without derating
Winding protection	3 x PT100 RTD in the stator winding

2.2 Bearings

Model	Drive end	Non-drive end
M-160L	6309 C3	6209 C3
M-200L	6312 C3	6212 C3
M-225M	6313 C3	6213 C3

3. Installation

3.1 Mounting

- Mount the motor on a flat, rigid base. Check for soft foot before tightening the feet: maximum 0.05 mm.
- Leave a free space of at least one quarter of the frame height behind the fan cover so that cooling air can enter.
- Do not mount the motor where hot exhaust air from other equipment can be drawn into the fan.

3.2 Alignment

Align the motor to the driven machine within 0.05 mm parallel offset and 0.05 mm per 100 mm angular. Belt drives must be tensioned according to the belt manufacturer; excessive belt tension overloads the drive end bearing.

3.3 Electrical connection

- Connect according to the diagram inside the terminal box. 400 V supply: delta connection for these frame sizes.
- The cable must be sized for the rated current and the installation method. Earth the motor at the earth terminal in the terminal box and on the frame.
- Set the overload relay to the rated current on the nameplate. Do not set it higher to avoid nuisance trips; find the cause instead.
- Connect the PT100 winding sensors to the protection relay.

3.4 Starting

The motors are designed for direct-on-line (DOL), star-delta or variable frequency drive (VFD) starting. With DOL starting, the starting current is about 7 times rated current.

Maximum number of starts: 3 starts per hour, evenly spaced, or 2 consecutive starts from cold. Frequent starting is a common cause of winding overheating because each start heats the rotor and stator.

4. Operation and protection

4.1 Temperature monitoring

The stator winding temperature is measured by the PT100 sensors. Typical winding temperature at full load and 40 °C ambient is 110 to 120 °C.

Winding temperature	Action
Up to 130 °C	Normal operation
130 °C	Warning: check load and cooling
145 °C	Alarm / trip: the protection relay stops the motor

Insulation life halves for every 10 °C of operation above the rated temperature of the insulation class. Repeated operation above 145 °C causes premature winding failure.

4.2 Vibration limits

M-Series motors are classified as medium machines on rigid foundations. Overall vibration velocity on the bearing housings:

Zone	Velocity (mm/s RMS)	Meaning
Good	below 1.8	New or recently serviced machine
Acceptable	1.8 to 2.8	Unrestricted operation
Warning	2.8 to 4.5	Plan maintenance
Alarm	above 4.5	Damage likely; investigate immediately

4.3 Supply quality

- Voltage tolerance: +/-5% of rated voltage at rated frequency.
- Voltage imbalance between phases should not exceed **1%**. An imbalance of 1% can cause a current imbalance of 6 to 10% and extra heating of the winding. Above 1% the motor must be derated.
- Loss of one phase (single phasing) while running causes a rapid rise in current and temperature; the overload relay must trip within seconds.

4.4 Current

The running current at full load must not exceed the rated current on the nameplate. A sustained current above 105% of rated current is a warning; above 115% the overload relay will trip.

5. Lubrication

The M-Series motors from frame 200 have regreasable bearings with grease nipples on both end shields. Frame 160 motors have sealed-for-life bearings.

5.1 Regreasing intervals at 1500 rpm

Bearing	Grease quantity	Interval
6312 C3	20 g	4,500 h
6212 C3	13 g	6,000 h
6313 C3	25 g	4,000 h
6213 C3	15 g	5,500 h

- Halve the interval for ambient temperatures above 40 °C or for vertical mounting.
- Use a polyurea or lithium complex grease, NLGI 2 or 3. The factory grease is a polyurea grease; check compatibility before changing grease type.
- Regrease with the motor running. Open the grease outlet plug and let the excess grease escape for about 1 hour before closing it.

5.2 Storage

If a motor is stored for more than 3 months, rotate the shaft by hand every month to prevent false brinelling of the bearings. Measure the insulation resistance before putting a stored motor into service (see M-ETP-05).